

THE LEDGER FRAMEWORK

Market Data Manifesto

Governing Principles for Market Data — Subordinate to the Architectural Constitution

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The record holds observations; every price beyond them is derived from them.

Abstract

The Ledger records observations, not prices. A market data fact enters the record exactly once, as a moveless observation admitted through the one door; every curve, surface, correlation, and calibration is a deterministic function of recorded inputs, of two kinds the document keeps apart. A *projection* computes over the record and rebuilds from it alone; a model's output *re-enters as a recorded observation* (C-14.9, C-14.15). Reading back any recorded value is unconditional; re-deriving a model's output is what needs the model. This manifesto states the principles that make append-only, event-driven, projected market data the only sensible way to hold market data inside the Ledger. Each principle is an *article*, derived from six of the Constitution's eight commitments — auditability, reproducibility, time travel, correctness, order, and simulability — and resting on the clause of the Constitution that carries it. The document is rated against the Constitution, never the other way round; where the two conflict, the Constitution prevails.

1. Market Data in the Ledger's Terms

A market data fact is an **observation**: a recorded statement, attributed to a source, that a named observable — a price, a rate, a fixing, an index level — took a value with respect to a world-moment. The observation is a fact about what a source said; it is not the truth about what a thing is worth. Any “true” price behind the observation is a modelling posit that lives in a model, never a claim the record makes.

The boundary is sharp. The source of an observation sits outside the trust boundary, and the Events Executor captures the arrival and records it (C-5.4). An observation enters the immutable log only as a moveless *observation-recording transaction* — moveless because it records data and moves no balances — admitted through the single Transaction Executor and folded into a *home*, its resting place in the record from which the contracts that need it read (C-4.8). (Throughout, to *fold* the log is to replay it in order, accumulating state; the current state is the initial state plus the fold of all observations.) There is no second door for market data: the one canonical record and the single writer extend, without exception, to every observation. A kind of market data is declared before any observation of it crosses — its fields, the market data operators that adjust it across events (C-9.2), and the grain of the cause-derived identifier — an identifier computed from what caused an arrival, so a redelivery of the same cause collides with it — that decides when two arrivals are the same fact, all registered on the record first, so an arrival of an undeclared kind is quarantined, never guessed (C-4.12).

Vocabulary, mapped once. This manifesto uses the Constitution's fixed names and coins

no synonym. Its prior art, the Market Data Calibration Manifesto, speaks a different dialect, mapped onto the Constitution’s vocabulary here once and not carried further:

- an *Oracle* is an external source; the framework has no “oracle” primitive and introduces none (Article MD-3);
- an *attestation* is an observation, recorded by an observation-recording transaction (C-4.8);
- *event time* is execution time; *knowledge time* splits into monitor time and door time (C-2.7);
- a *posterior* over parameters is a derived object (a value computed from recorded inputs; MD-6): a projection where its recipe computes over the record, a re-entered observation where it runs a model (C-14.15);
- a *model configuration* is the recorded, versioned terms of a derived object (C-4.4, C-7.2).

The Constitution’s own term *the market data operator* names one thing only — the corporate-action transform of C-9.2 that carries pre-event data into the post-event frame — and is never used for a source.

2. The Articles

Each article derives from one or more of the six commitments and rests on a clause of the Constitution. The identifiers MD-*n* are append-only: never reused, never renumbered, so later documents may cite them.

MD-1. The observation is the atom of market data. A market data fact is an observation, ^{MD-1} and an observation is recorded exactly once. It enters the immutable log as a moveless observation-recording transaction, admitted through the single writer, and folded into a home. “Exactly once” is a mechanism, not a wish: each arrival carries its cause-derived identifier, and an arrival identical under it is *absorbed* as a duplicate, not recorded again (C-2.7, C-12.3, C-14.3). Absorption is not a silent drop: no new value is recorded, but the fact of the redelivery — its arrival provenance and monitor time — is retained, so a healthy absorb is distinguishable from a feed that has gone quiet. Three arrivals that look alike must be told apart, and the identifier’s grain — a registered property of the data kind (§1) — is what tells them: the same print *redelivered* is absorbed; a vendor *correction* is a distinct, later observation that names the value it corrects (MD-10); a genuinely *new* print at the same instant is a second observation, recorded in its own right. A grain too *fine* records a mere redelivery twice and double-counts; a grain too *coarse* absorbs a genuine second print and loses it.

That the registered grain is correct is a trust assumption, named beside TA-ARRIVAL. The double-count is loud — the extra observation is on the record to be found — but the over-coarse loss is *silent*: the second print is treated as a duplicate and never recorded, and where a feed delivers two distinct prints identical under every field it sends, no grain can separate them. This is the one residual case of silent loss, and the manifesto does not pretend a grain always exists to prevent it; it is caught where TA-ARRIVAL’s gaps are caught — at the perimeter, by reconciling arrival counts against recorded observations. Subject to that, nothing about market data is written twice: the value a source stated is its sole canonical copy, and every price, curve, and surface that mentions it is computed from it, never stored beside it.

(*Auditability, Correctness; C-1.4, C-4.8, C-2.7, C-12.3, C-14.3.*)

MD-2. Capture precedes judgement. The record admits an observation before anything ^{MD-2} decides whether to believe it. A stale, fat-fingered, or off-consensus quote is well-formed: it enters through the normal door as an observation-recording transaction (C-4.8), and judgement is deferred because economic correctness is never gated at admission — it is detected after the fact by recomputation (C-13.2, C-13.3). A quote that is not even processable — a corrupted payload,

an unresolvable reference — is still not lost: it is captured with the provenance of its arrival and stands as a visible open item (C-4.12). Either way the Ledger captures, then classifies; it never refuses to record in order to reject. Whether an observation is used, down-weighted, or ignored is a separate and later question, answered by a derived object over the record, never by the door. A “rejected” observation is one a calibration assigned no weight — present, inspectable, and open to a different classification by a later derivation — not one the record never held. Capture-then-classify is a *principle*, and its guarantee runs *from arrival*: whether an implementation’s pipe can hold the full firehose at the open is a capacity question outside this document (trust assumption TA-ARRIVAL) — but the principle forbids using capacity as a licence to drop silently. What the boundary cannot admit is a named, visible residual; what never reaches the boundary is the perimeter’s concern, reconciled there, never a fact the record can claim (C-4.12).

(*Auditability, Order, Correctness; C-4.8, C-13.2, C-13.3, C-4.12.*)

MD-3. The source is untrusted; its delivery is a trust assumption. The origin of an observation — an exchange, a broker, a polling service, a fixing provider, an on-chain feed — is machinery beyond the system’s trust boundary: an external source, not one of the system’s own machines (C-5.1, C-5.8), and the framework grants no source the standing of truth. That an observation reaches the boundary at all is named as a trust assumption and reconciled against external authorities at the perimeter; it is never a fact the fold may assume (C-4.12). The framework therefore has no privileged “oracle” component and introduces none: a source is where an observation came from, recorded as provenance, and nothing more.

(*Auditability, Correctness; C-5.1, C-5.4, C-5.8, C-4.12.*)

MD-4. Every observation bears three times, and as-of and as-at are two of them. By the Constitution an event carries an execution time — the world-moment its source asserts it happened, the time a court would enforce, contestable only in the world and corrected only by a later event; a monitor time — when the boundary observed it; and a door time — when the single writer admitted it (C-2.7). Bitemporality is not a feature bolted onto market data; it is these three times, already present. The *as-of* coordinate of an observation — the world-moment its value concerns — is its execution time; the *as-at* coordinate — when the record came to hold it — is its door time; monitor time is the further provenance that splits an observation’s lateness into the world’s delay and ours. Because both coordinates are recorded, the system computes both honest answers at any moment: what was known as of a past date, and what is known now about that date (C-12.1).

(*Time travel, Order; C-2.7, C-12.1.*)

MD-5. Order is meaning, and the authoritative fold is execution-ordered. Observations are events, and their sequence is load-bearing: a derived object depends on which observations preceded it — a surface fitted before a shock is not the surface fitted after it. Meaning lives in execution order. An observation that arrives late, whose execution time precedes observations already folded, takes its place among them; every **projection** downstream refolds to include it (C-2.7), while a **re-entered observation** whose input window it lands in cannot re-fold itself — the ledger runs no model (C-14.9) — and is instead flagged stale for re-derivation (MD-8). Neither is silent: the insertion is flagged, every changed or staled value is flagged, and the difference is published as a named explain item — a line in the profit-and-loss explain attributing the change to its cause (C-12.6). The as-known view — what the book showed with only the observations it then held — survives unrewritten beside the corrected one.

(*Order, Time travel; C-2.7, C-12.6, C-14.9.*)

MD-6. Every derived object is a deterministic function of recorded inputs. A curve, a surface, a correlation, a calibration is a function of what the record holds — the observations it

consumes, any derived objects it consumes, each pinned to its own version and *cut* (the exact as-at boundary that fixes which observations were in force; MD-12), the recipe that defines it, and, where the recipe draws on randomness, a recorded seed. Two kinds of derived object must be told apart, and the Constitution names both.

Where the recipe computes *over the record* — a discount factor read off recorded quotes, an adjusted value from the market data operator — the object is a **projection**: it stores nothing and rebuilds from the record alone, by any party, today or at a past date (C-2.2, C-3.2, C-14.11). Where the recipe *runs a model* — a filter, an optimiser, a fitted surface — the ledger does not run it (C-14.9); the model is evaluated outside the fold and its output re-enters through the one door as a new **observation** (C-14.15). That number is then stored, like any observation, and two senses of getting it back must not be confused: *reading it back* is always possible from the record alone, because the value is on the record; *re-deriving* it — showing the number follows from its inputs — additionally needs the retained model and the numerical environment it ran in, a governance matter outside this scope. So a re-entered observation is read back unconditionally and re-derived only with the model.

Either kind carries, as recorded provenance, a **complete lineage**: the resolved input cut — every observation and every derived object it consumed, each pinned to version and cut — and, for a model output, the model version. Lineage reaches through the whole chain to the observations at its leaves, so what an object depends on is always legible from the record. A derivation that *fails* is not an absence either, but which kind of fact it is follows the same split. A **projection** that fails — a bootstrap with too few quotes at this cut — is itself a projection: its failure is legible by recomputation, never stored as a standing fact, and refreshes to success if a late arrival fills the gap (MD-5). A **model run** that fails — a fit that does not converge — re-enters as a recorded observation carrying its diagnostics and lineage, and is itself subject to staleness if its inputs are later corrected. A failed *capture* (MD-2) is stored for a different reason: it is an input-side arrival, not an output-side derivation. In every case an absent object never passes for one never attempted. The demand is on the inputs, not the engine: record enough — observations, derived inputs, recipe, seed, and the lineage that ties them — that the object is a deterministic function of the record.

(*Reproducibility, Simulability, Correctness; C-2.2, C-3.2, C-14.9, C-14.11, C-14.15.*)

MD-7. The prior is a term; the posterior is a derived object. A Bayesian calibration is MD-7 not banned; it is placed. The prior — including any assumed dynamics for how a parameter moves between observations — is a declared, recorded term. The posterior is a derived object of that prior-term, the recorded observations, and the recorded seed (MD-6): a projection where it is a closed computation over the record, a re-entered observation where a filter or sampler evaluates a model. Either way it is *not* stored state — no “current posterior” sits beside the record and updates in place; the maxim that today’s posterior is tomorrow’s prior names a recomputation over the log, not a mutable variable. The framework mandates no dynamics — a random walk, a martingale, a mean reversion are modelling choices a recipe may declare, never axioms the framework asserts — and whatever is declared is recorded, so the posterior is reproducible on MD-6’s terms.

(*Correctness, Reproducibility; C-2.8, C-4.11, C-14.15.*)

MD-8. A broken state cannot hide. A recurring pathology of calibrated systems is the stored MD-8 parameter that has drifted from what the information implies: the system knows something it has not yet written into its numbers. The Ledger defeats it, but the mechanism differs by kind of derived object, and honesty requires naming both. A **projection** stores nothing — recomputed from the record on every read (C-4.11) — so it cannot lag the record at all: the drifted-parameter state is unrepresentable, and under a joint recipe an on-demand read is the

correct joint conditional expectation, so even two correlated legs cannot hold a divergent pair. A **re-entered observation** does store a number: a fitted surface, or a joint vol-and-correlation fit for a spread, sits on the record because the ledger runs no model (C-14.9). It cannot drift in place — a new fit is a new observation, forward, naming the inputs it consumed (MD-6), never an edit of the old — so the system never quietly overwrites what it knew. But it can fall out of date when an input it consumed is later corrected or superseded (MD-10, MD-5), and there the ledger does not silently recompute it. Instead its staleness is a recorded fact: the recorded lineage shows the input moved under it, and the stale fit stands as a flagged open item awaiting re-derivation. So the joint spread surface cannot silently disagree with a corrected leg — not because it re-fits itself, but because its disagreement is visible. So this visibility is reached by default: a consumer reads a re-entered observation through a projection — “the current fit for X”, which selects the latest non-superseded observation and carries its staleness flag — never the raw stored value, so an ordinary read inherits the check. What the framework forbids everywhere is the *silent* divergence: for a projection it cannot arise; for a re-entered observation it cannot hide.

(*Correctness; C-1.5, C-4.11, C-11.5, C-14.9.*)

MD-9. Constraints live in the derived object; no-arbitrage is declared, not assumed. MD-9

A recorded observation carries no promise of mutual consistency — a stale quote and a fresh one may cross, and two sources may disagree; the record holds both. No-arbitrage, smoothness, and monotonicity are constraints a derived object imposes on its own output, not properties the record must have. A derived object that cannot meet the constraints it declares records that it cannot — a visible diagnostic — rather than failing to exist or, worse, serving an inadmissible object as though it were sound. No-arbitrage is not negotiable for the object that claims it, but it is enforced by detection, not by admission: an arbitrageable surface served as arbitrage-free is an economic error, caught after the fact by recomputation, not prevented at a door (C-13.2, C-13.3). The discipline is that the claim and its test are both on the record. Which no-arbitrage conditions are correct, and whether a fitted object hedges, are matters of model choice and lie outside this scope (C-Scope.11): the manifesto mandates that a derived object’s constraints be recorded and their satisfaction be a recorded diagnostic, not which constraints a modeller should choose.

(*Auditability, Correctness; C-13.2, C-13.3, C-Scope.11.*)

MD-10. Corrections repair forward; the original is never overwritten. MD-10

A market data fact found wrong is not edited. The discovery is a new recorded observation that names the wrong one, and the discrepancy stands as a visible open item. What happens downstream splits by kind: every **projection** over the corrected fact recomputes automatically on the next read; a **re-entered observation** that consumed the wrong fact cannot re-run itself (C-14.9), so its recorded lineage marks it stale and it stands as a flagged open item until a fresh fit supersedes it forward (MD-8). A bulk retraction — a vendor voiding a whole day — is many corrections at once; the explain may summarise them as one named item rather than flood the book, so the signal is not lost in the flags. Where money already moved on the strength of the wrong fact, it moves back only as an authorised compensating transaction (C-12.4). The observed value stays on the record exactly as observed. Adjustments a later event forces on earlier data — a split rescaling a price, a dividend rescaling a cash-per-share figure — are computed at each read by the market data operator (C-9.2, C-9.3), which is intrinsic to the pairing of the kind of data and the kind of event, declared once and never improvised. The original observation is preserved; the adjusted value is derived; projections are recomputed from operator-adjusted inputs.

(*Auditability, Time travel, Correctness; C-9.3, C-12.4, C-12.6, C-14.9.*)

MD-11. Simulated market data is real market data under a different seed. MD-11

of the market data universe is the initial state plus the fold of all recorded observations (C-2.8). A stress scenario, a backtest, a Monte-Carlo path is not a separate system but the same machine folding a generated stream of observations from a recorded seed; production is simply the one path whose observations happened to be real. Because the seed is the single non-record input and it too is recorded, every simulated path replays exactly, and a calibration run over simulated observations is reproducible on the same terms as one run over real ones. The seed suffices as the sole non-record input only because every model output along the path is itself captured as a re-entered observation (MD-6); a path that ran a model but recorded only the seed could not replay without that model.

(*Simulability, Reproducibility; C-2.8, C-10.1.*)

MD-12. Derivation composes. A derived object may consume another derived object — a volatility calibration reads a discount curve, a spread model reads two surfaces. When it does, the dependency — which object, at which version and cut — is itself part of the recorded recipe (MD-6), so the composite is again a deterministic function of the record, and the split of MD-6 composes with it. Along a chain of **projections**, reconstruction runs the whole chain from the record, bit-for-bit. Where the chain passes through a **re-entered observation**, that node enters as a *leaf*: its recorded value is read back from the record, and re-deriving it needs its retained model and numerical environment (MD-6) — so reconstruction of the composite is bit-for-bit given those, and read-back always. Either way, because each input is fixed at its own recorded cut, never read from a mutable store, the “mid-update input” cannot arise (MD-8): a curve feeding a calibration is consistent with that calibration by construction, not by scheduling.

(*Reproducibility, Correctness; C-2.2, C-14.11.*)

3. The Life of an Observation

One observation can walk the whole picture, and the articles are easier to hold once it has. Follow AAPL’s official close on a Tuesday, needed by an equity volatility surface.

- **Recorded once.** The exchange, an external source, publishes AAPL’s Tuesday close of 150.20; the Events Executor captures it and the Transaction Executor admits a moveless observation-recording transaction. The value 150.20 is now on the record, once; a redelivery of the same print is absorbed, not recorded again (MD-1, MD-3).
- **Three times.** The observation carries its execution time (Tue 16:00, the world-moment the close concerns — its as-of coordinate), its monitor time (when the boundary saw it), and its door time (Tue 18:30, when it was admitted — its as-at coordinate) (MD-4).
- **Read back, not drifting.** Tuesday’s surface consumes 150.20. If it is a projection it rebuilds from the record on read; if it is a fitted model it re-enters as a recorded observation pinned to the closes it consumed (MD-6). Either way its value is always readable back from the record; re-deriving a fitted surface needs its retained model. No surface is kept as a mutable copy that drifts (MD-8).
- **Corrected forward.** On Thursday the exchange restates the Tuesday close to 150.50. This is a correction, not a late arrival: a new observation naming the old one, sharing Tuesday’s execution time, separated only by door time (Thu) (MD-10). The original 150.20 stays on the record exactly as observed.
- **Two honest answers, two numbers.** Now the daily question. *Tuesday’s surface as published on Tuesday* was built at the cut as-at Tue 18:30 — it used 150.20, and that view survives unrewritten. *Tuesday’s surface as recomputed now* is built at the cut as-at today — it uses the corrected 150.50. Same as-of date, two as-at cuts, two different surfaces; the cut

is the coordinate that selects which (MD-4). A projection yields both by recomputation; a fitted surface yields the first as the observation recorded then and the second only once re-fitted, its staleness flagged until it is (MD-8).

- **Late arrival.** Suppose instead a Tuesday-stamped trade print arrives only Thursday. Its execution time places it among Tuesday's data, so every projection over that window refolds to include it and publishes a named explain item, while a fitted surface for that window is flagged stale for re-derivation (MD-5). A value learned Thursday changes what Tuesday's book should have said.
- **Simulated the same way.** Replace Tuesday's real close with a generated one from a recorded seed and the same machine runs unchanged; the surface reproduces on identical terms (MD-11).

4. What This Manifesto Does Not Govern

This manifesto governs how market data is recorded, ordered, corrected, and projected. It does not govern:

- **price formation and price discovery** — where a price comes from in the market is not a question the record answers;
- **which model, prior, or no-arbitrage condition is correct** — that is model choice, and its test is hedging performance, outside this document;
- **model governance, validation, and the retention of models** — a value reproduces from recorded prices unconditionally, a model's number only if the model is kept, and keeping it is governance;
- **settlement** — the record projects committed activity toward it but does not perform it.

These lie outside the Ledger boundary and are reconciled against it only at its perimeter (C-Scope.11).

5. Subordination

This manifesto is subordinate to the Architectural Constitution and is rated against it, never the other way round. Its vocabulary is the Constitution's, one name per component, and it introduces no synonym for any fixed term. Where a principle here would conflict with the Constitution, the Constitution prevails, and the conflict is parked with the exact text of the proposed amendment and the exact clause it would replace — never resolved by quiet narrowing. A departure from this document, once genuine, is made by amending it explicitly first; an article identifier is never reused and never renumbered.

Conclusion

Reconciliation of market data — one desk's surface disagreeing with another's, today's mark disagreeing with the mark reproduced tomorrow — has the same root cause as every other reconciliation: the same fact stored more than once. The Ledger removes it here as everywhere. An observation is recorded once; every curve, surface, and calibration is a deterministic function of what the record holds; corrections repair forward and originals survive; and the whole market data universe is the initial state folded over its recorded observations. Operate market data any other way inside the Ledger and you reintroduce the divergence the architecture exists to remove. Operate it this way and there is nothing left to reconcile.